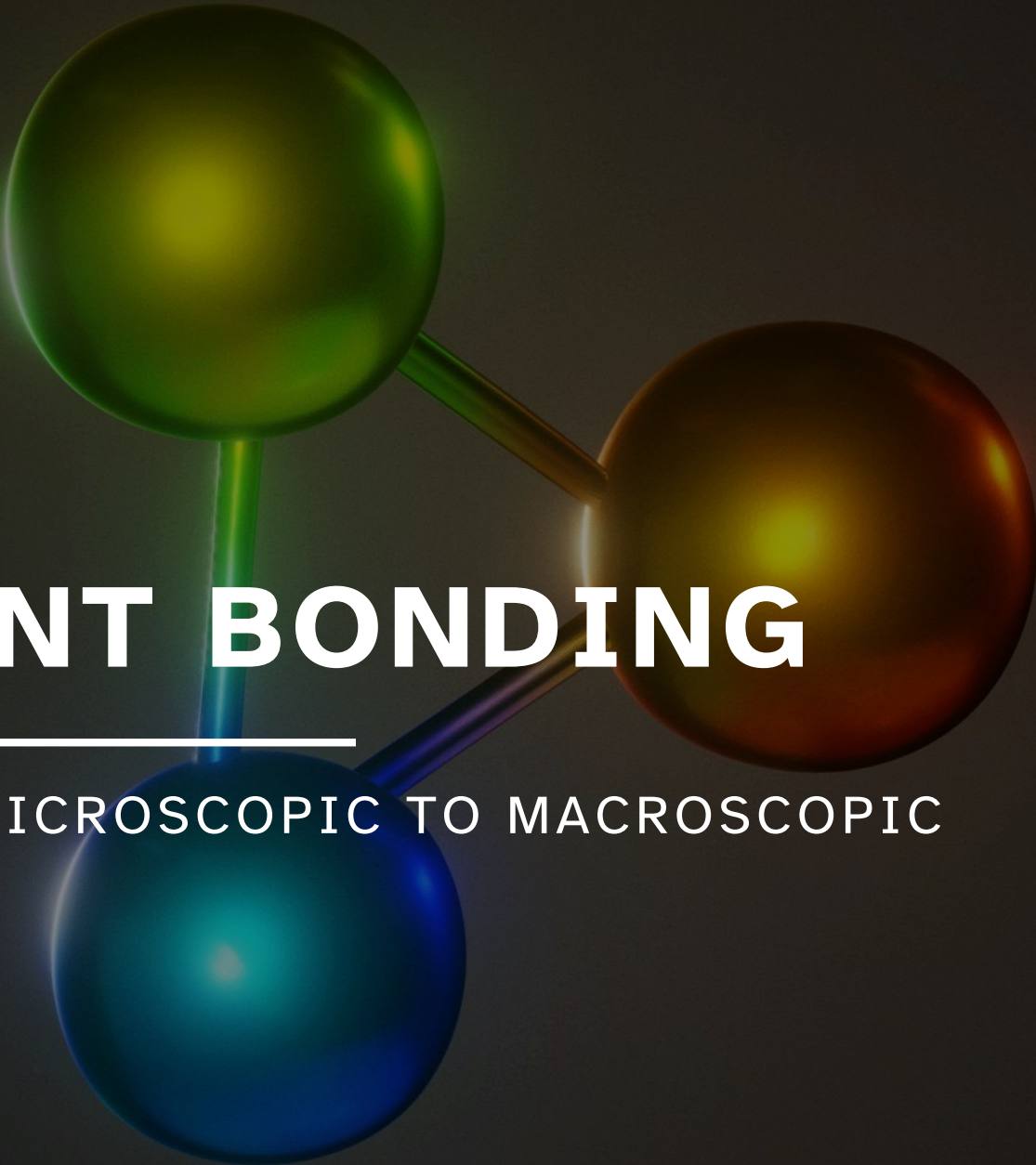




# 10B COVALENT BONDING

---

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

# LEARNING OBJECTIVES

## CHAPTER 10

1. Predict the number of bonds an atom is likely to form in a molecule.
2. Draw Lewis structures for molecules that follow the octet rule.
3. Define and apply electronegativity and its periodic trends.
4. Draw Lewis structures for molecules for which multiple valid Lewis structures exist (called resonance).
5. Draw Lewis structures for molecules that do not obey the octet rule by application of the typical exceptions exhibited by H, Be, B, and expanded octets.
6. Determine bond polarity using electronegativity values.
7. Describe and apply the relation between bond length and bond strength.
8. Apply bond enthalpies to calculate enthalpies of reaction.

# Exceptions to the Octet Rule

10.4

*Limitations/Drawbacks to Lewis structure theory and “bandage solutions”*

- 1) Electrons are treated as even pairs of electrons: 2, 4, 6, 8
  - What about molecules with an *odd number of electrons*?
  - What about radicals (molecules with *unpaired electrons*)?
  - We will not deal with these cases.
- 2) Resonance: multiple Lewis structures needed
- 3) Incomplete octet: central atom with <8 electrons
- 4) Expanded octet: central atom with >8 electrons

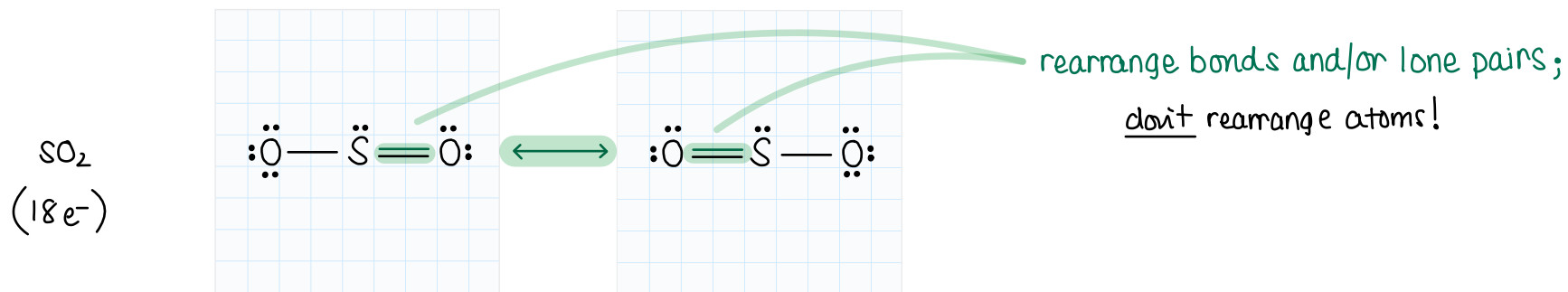
Let's work through these more “exotic” molecules' Lewis structures!

# Resonance and Electronegativity

10.3 (we are not covering formal charges)

In some cases, a molecule cannot be represented accurately by a *single* Lewis structure but rather by *multiple valid* Lewis structures.

- These molecules/ions are said to exhibit resonance.
- The individual Lewis structures are called resonance structures or forms.
- These structures differ *only* in the placement of the double bonds, triple bonds, and/or lone pairs.



# Resonance and Electronegativity

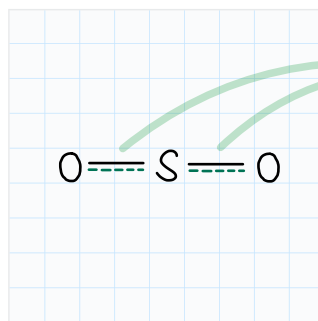
10.3 (we are not covering formal charges)

Misconceptions: The individual resonance structures do not exist. The double-headed arrow does not mean that the molecule alternates between the resonance structures.

The *real/true structure* of the molecule is called the resonance hybrid.

- It is constructed by taking a *geometric average* over all resonance structures.

note: this is HARD to draw and think about,  
which is why we draw multiple EASY  
resonance structures instead.



the two bonds are equivalent;  
neither is a single bond (—);  
neither is a double bond (==);  
both are somewhere in between (====); *weird!*

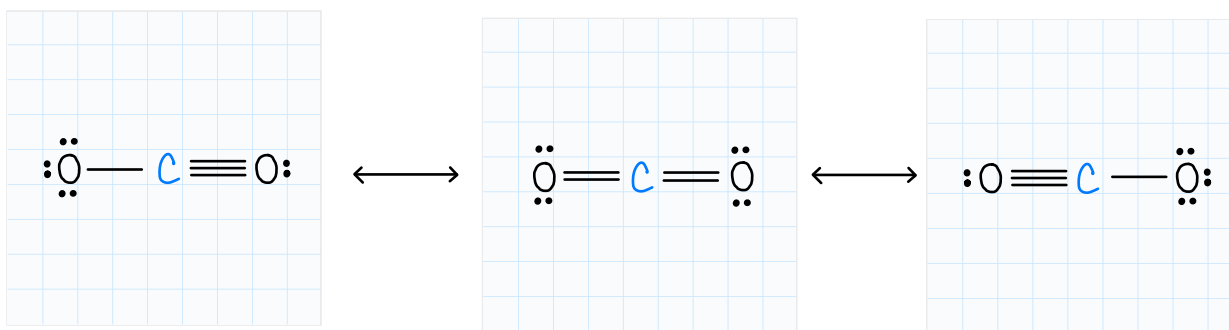
- The pairs of electrons involved in the resonance structures are said to be “spread out” among the different positions: *delocalized bonds!*

# EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.4

Draw three *different* Lewis structures for  $\text{CO}_2$ .

$4e^- + 2(6e^-) = 16e^-$  total



note: 2<sup>nd</sup> structure is best for formal charge reasoning but we won't consider this

1. Count total # valence  $e^-$
2. Pick central atom
3. Connect to central atoms using single bonds
4. Satisfy octet for outer atoms, or duet for H
5. Satisfy octet for central atom
  - a) If leftover  $e^-$ , assign as lone pairs
  - b) If no  $e^-$  left over, move lone pairs from outer atoms to double/triple bonds
6. Verify octet/duet rule and correct # total  $e^-$

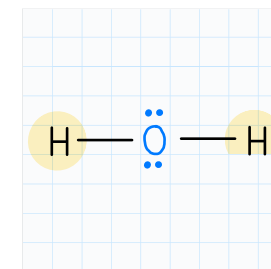
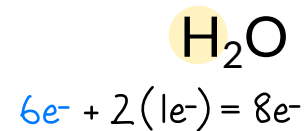
Periodic table showing the elements Carbon (C), Nitrogen (N), and Oxygen (O) highlighted in blue and circled, indicating their relevance to the problem.

# Exceptions to the Octet Rule

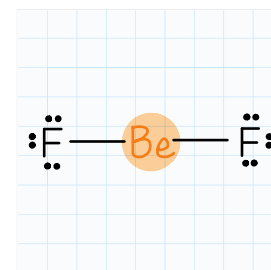
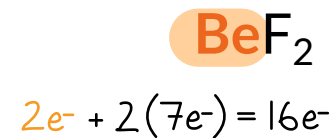
10.4

## Incomplete octets

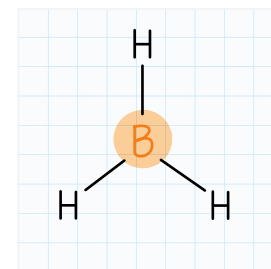
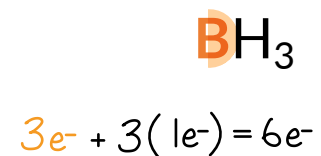
- **H** and **He**: only needs  $2 e^-$  (recall duet rule)



- Beryllium, **Be**: okay with  $4 e^-$



- Boron, **B**: okay with  $6 e^-$



# Exceptions to the Octet Rule

10.4

Q: How to know if needed?

A: Look at total e- count!

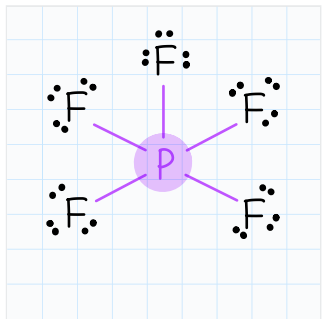
## Expanded octets

- The **central atom** can expand or exceed the octet rule **IF it must**.
  - The central atom must be from **periods/rows:  $n=3-7$**
  - Never expand the octets for outer atoms.

obvious



$$5e^- + 5(7e^-) = 40e^-$$

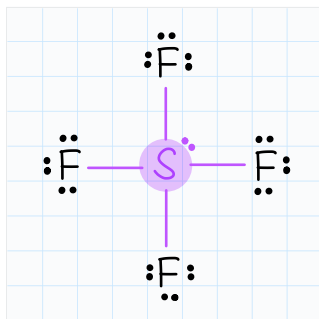


$12e^-$

not as obvious



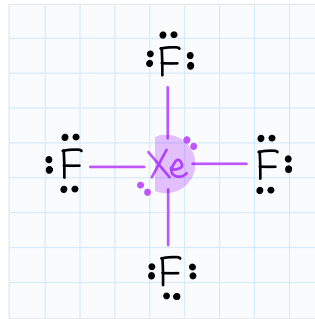
$$6e^- + 4(7e^-) = 34e^-$$



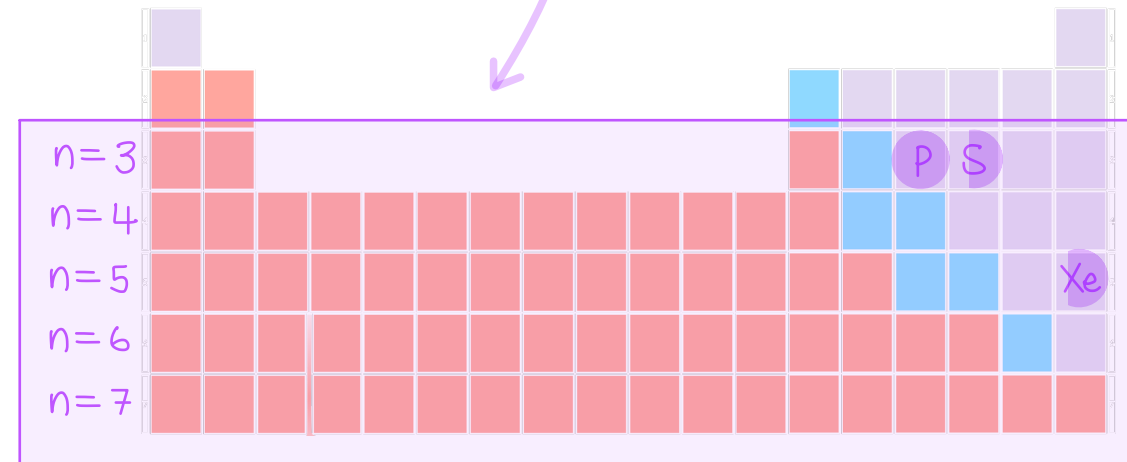
$10e^-$



$$8e^- + 4(7e^-) = 36e^-$$



$12e^-$

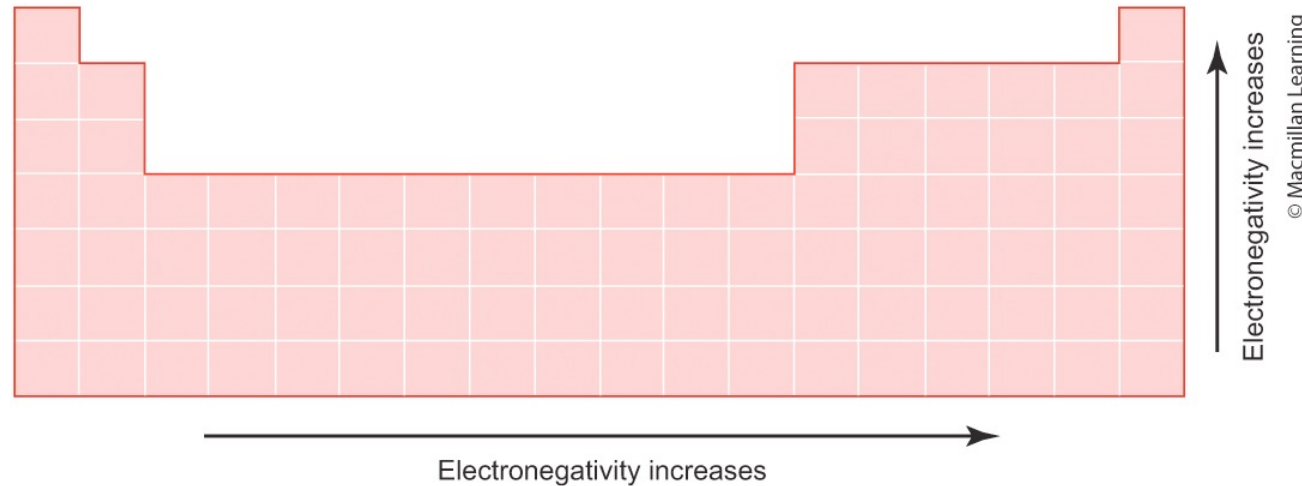


# Resonance and Electronegativity

10.3 (we are not covering formal charges)

## Electronegativity (EN)

- The ability of an atom to attract *shared* electrons to itself in a bond.



**FIGURE 11.15 General electronegativity trend**

- Different numerical scales exist, but Pauling scale is most common.

# Polar Bonds and the Bonding Continuum

10.5

The difference in electronegativity ( $\Delta EN$ ) between two atoms in a bond can be used to classify bonds by ionic character on a continuum.

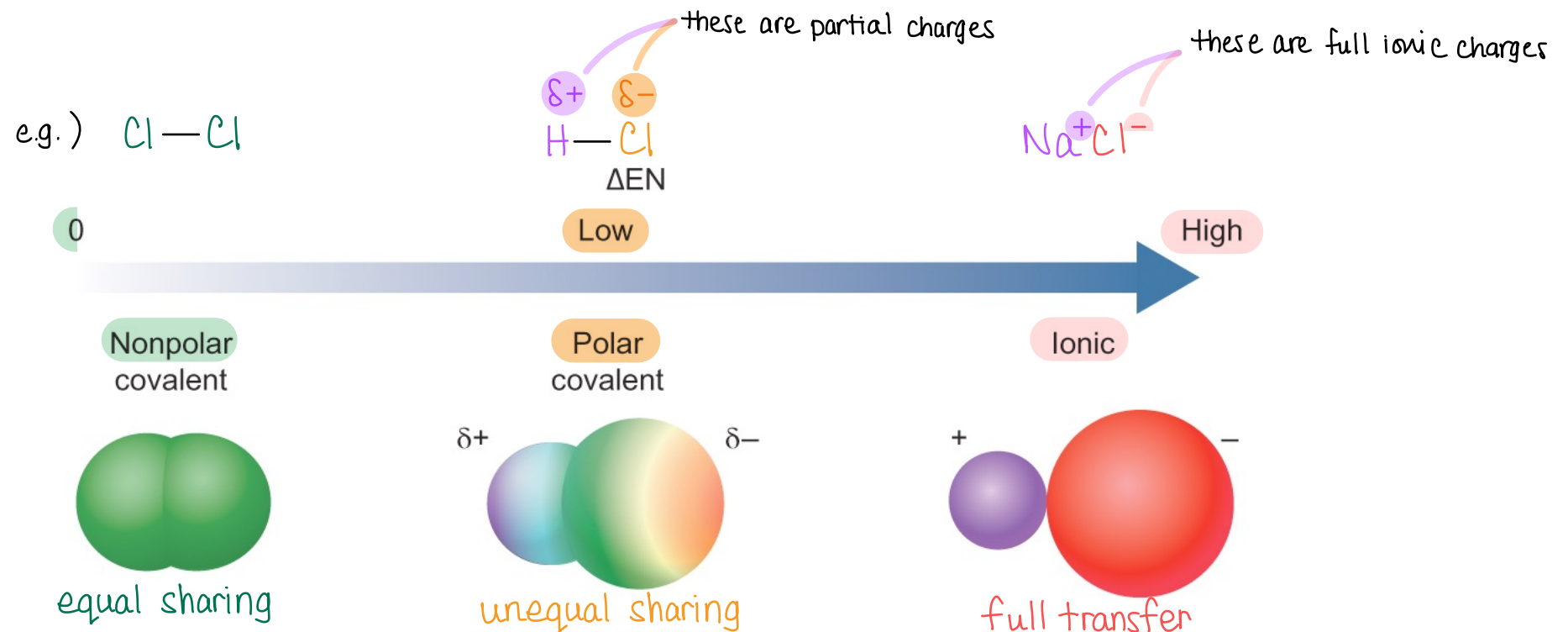
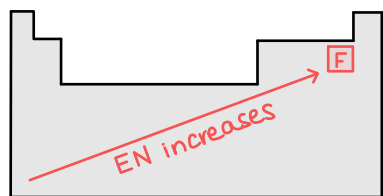


FIGURE 10.17 The bonding continuum



# Polar Bonds and the Bonding Continuum

10.5

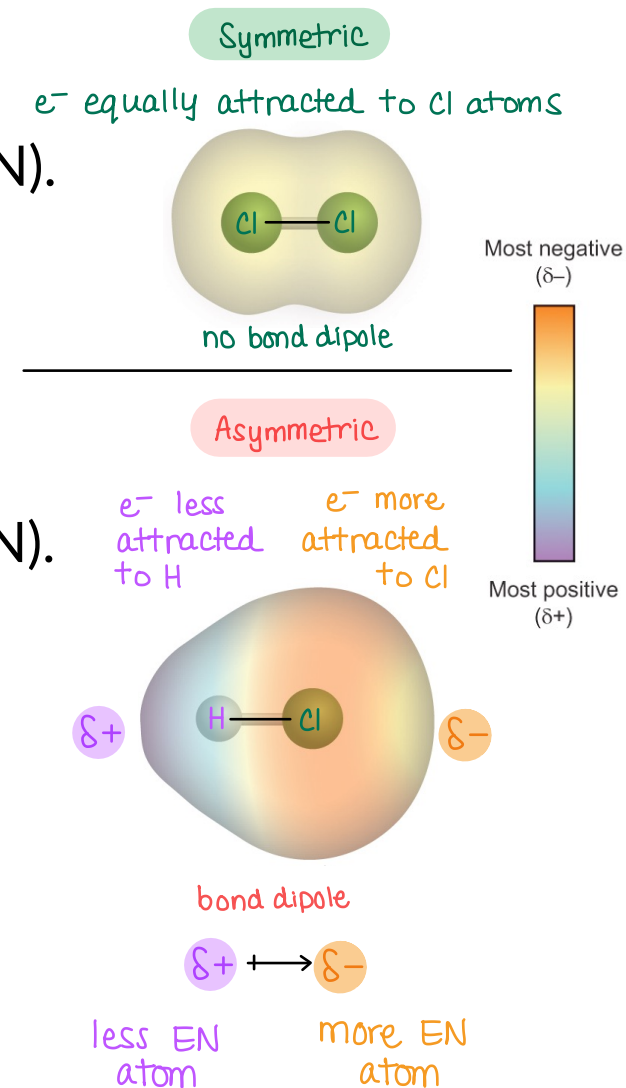
## Nonpolar covalent bond $:\ddot{\text{Cl}}-\ddot{\text{Cl}}:$

- Bond between two atoms of similar electronegativities (EN).
- The electrons are shared equally.

## Polar covalent bond $\text{H}-\ddot{\text{Cl}}:$

- Bond between two atoms of different electronegativities (EN).
- The electrons are shared unequally.
- There is a **partial positive charge ( $\delta+$ )** on the *less* EN atom.
- There is a **partial negative charge ( $\delta-$ )** on the *more* EN atom.
- The  $\delta+$  and  $\delta-$  constitute a bond dipole.

(vector;  $\longleftrightarrow$ )

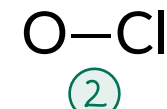
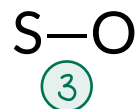


# IN-CLASS QUESTION 1

Learning Objective(s): 10.6

1 → 4

Rank the bonds in order of *increasing* ionic character.



|           |           |           |           |           |           |           |          |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|
| 1<br>1A   |           |           |           |           |           |           | 18<br>8A |
| H<br>2.1  | 2<br>2A   |           |           |           |           |           | He       |
| Li<br>1.0 | Be<br>1.5 | 13<br>3A  | 14<br>4A  | 15<br>5A  | 16<br>6A  | 17<br>7A  | Ne       |
| Na<br>0.9 | Mg<br>1.2 | B<br>2.0  | C<br>2.5  | N<br>3.0  | O<br>3.5  | F<br>4.0  | Ar       |
| K<br>0.8  | Ca<br>1.0 | Al<br>1.5 | Si<br>1.8 | P<br>2.1  | S<br>2.5  | Cl<br>3.0 | Kr       |
| Rb<br>0.8 | Sr<br>1.0 | Ga<br>1.6 | Ge<br>1.8 | As<br>2.0 | Se<br>2.4 | Br<br>2.8 | Xe       |
| Cs<br>0.7 | Ba<br>0.9 | In<br>1.7 | Sn<br>1.8 | Sb<br>1.9 | Te<br>2.1 | I<br>2.5  | Rn       |
| Fr<br>0.7 | Ra<br>0.9 | Tl<br>1.8 | Pb<br>1.9 | Bi<br>1.9 | Po<br>2.0 | At<br>2.2 |          |

Electronegativity increases

EN increases left to right : N < O < F ; S < Cl

EN increases bottom to top : S < O ; Cl < F

Easiest to use actual  $\Delta EN$  values

N—Cl  $\Delta EN = 3.0 - 3.0 = 0$

O—Cl  $\Delta EN = 3.5 - 3.0 = 0.5$

S—O  $\Delta EN = 3.5 - 2.5 = 1.0$

S—F  $\Delta EN = 4.0 - 2.5 = 1.5$

increasing  $\Delta EN$   
increasing ionic character

# Where did we start and where are we going?

Chemistry in Context

What is our overall goal?



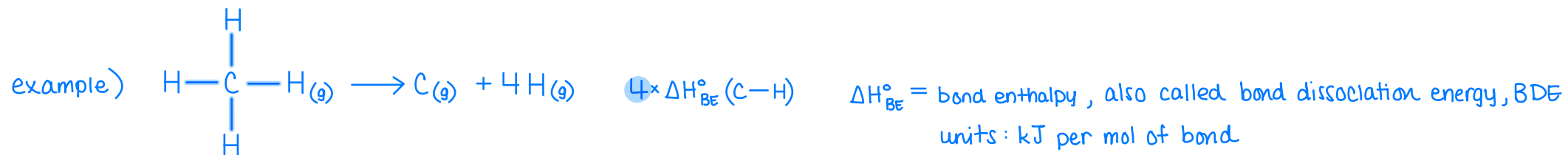
# Bond Enthalpy

10.6

## Bond enthalpy or bond energy ( $\Delta H_{BE}$ )



- The enthalpy change associated with breaking a specific bond in 1 mol of gaseous molecules.



note:  $\Delta H_{BE}^{\circ}$  are usually averaged over many different kinds of molecules with X-Y bonds

- Bond breaking is always *endothermic*, so all  $\Delta H_{BE} > 0$  (positive values).
- Stronger bonds are more stable and have larger bond enthalpies ( $\Delta H_{BE}$ ).

# Bond Enthalpy

10.6

## General trends

- Polar bonds tend to be stronger than nonpolar bonds.
- Multiple bonds (= and  $\equiv$ ) are stronger than single bonds (—).
- Stronger bonds tend to be shorter.
- Weaker bonds tend to be longer.

| Bond         | # of Electrons | Bond Enthalpy ( $\Delta H_{\text{BE}}$ ) | Bond Length |
|--------------|----------------|------------------------------------------|-------------|
| C—C          | 2 $e^-$        | 348 kJ/mol                               | 154 pm      |
| C=C          | 4 $e^-$        | 614 kJ/mol                               | 134 pm      |
| C $\equiv$ C | 6 $e^-$        | 839 kJ/mol                               | 120 pm      |

# Bond Enthalpy

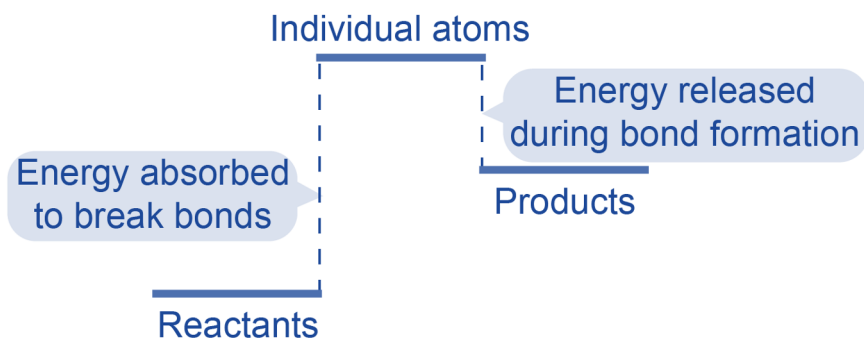
10.6

We can use bond enthalpies to contextualize/understand  $\Delta H_{\text{rxn}}$ .

- Bond *breaking* is *endothermic*  $\text{X}-\text{Y}_{(g)} \longrightarrow \text{X}_{(g)} + \text{Y}_{(g)} \quad \Delta H_{\text{BE}}^{\circ} > 0$
- Bond *formation* is *exothermic*  $\text{X}_{(g)} + \text{Y}_{(g)} \longrightarrow \text{X}-\text{Y}_{(g)} \quad -\Delta H_{\text{BE}}^{\circ} < 0$

ENDOTHERMIC RXN  $\Delta H_{\text{rxn}} > 0$

- strong bonds broken
- weak bonds formed



EXOTHERMIC RXN  $\Delta H_{\text{rxn}} < 0$

- weak bonds broken
- strong bonds formed

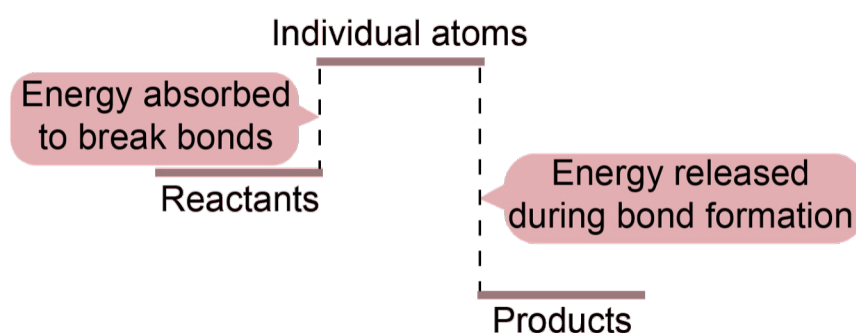


FIGURE 10.19

# Bond Enthalpy

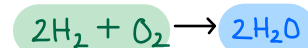
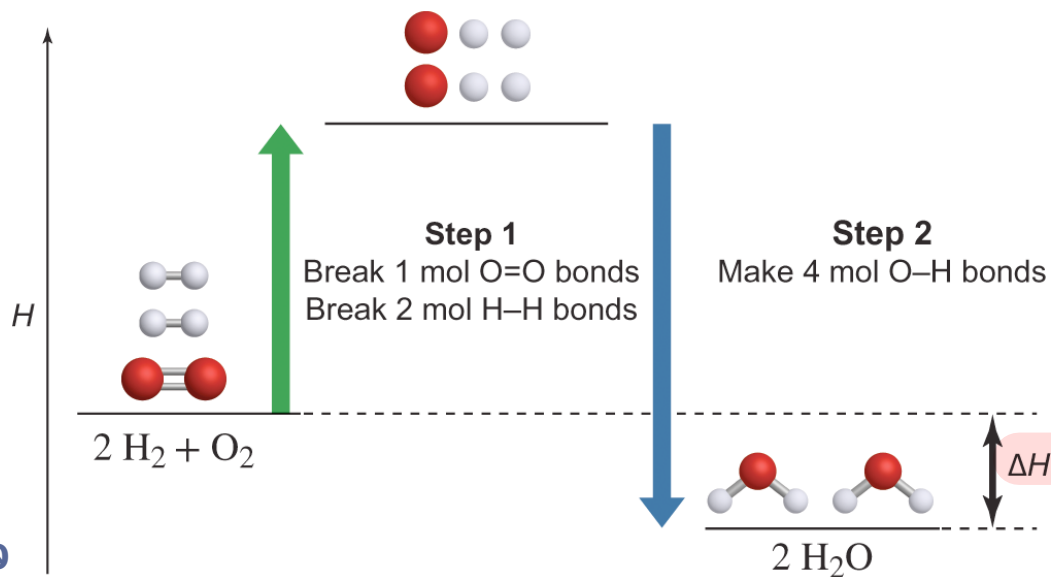
## 10.6

In some cases, we can even use bond enthalpies to *estimate*  $\Delta H_{\text{rxn}}$ .

- Difference between energy put into breaking bonds in reactants and energy released when forming new bonds in the products is *approximately*  $\Delta H_{\text{rxn}}$ .

$$\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{BE}}^{\circ} \left( \begin{array}{c} \text{all bonds broken} \\ \text{in reactants} \end{array} \right) - \sum \Delta H_{\text{BE}}^{\circ} \left( \begin{array}{c} \text{all bonds formed} \\ \text{in products} \end{array} \right)$$

subtraction explicit so don't put any extra negative signs  
i.e. all  $\Delta H_{\text{BE}}^{\circ}$  used in this equation are positive



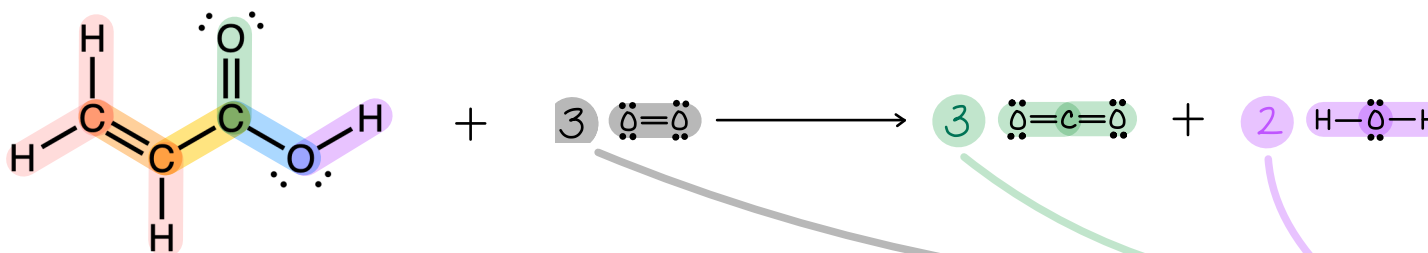
$$\Delta H_{\text{rxn}}^{\circ} = \left[ 1 \times \Delta H_{\text{BE}}^{\circ} (\text{O}=\text{O}) + 2 \times \Delta H_{\text{BE}}^{\circ} (\text{H}-\text{H}) \right] - \left[ 4 \times \Delta H_{\text{BE}}^{\circ} (\text{O}-\text{H}) \right]$$

FIGURE 10.20

# IN-CLASS QUESTION 2

Learning Objective(s): 10.12

What is the standard change in enthalpy ( $\Delta H_{\text{rxn}}^\circ$ ) for the combustion of acrylic acid ( $\text{C}_3\text{H}_4\text{O}_2$ )?



note: don't forget stoichiometry!

| Bond | Bond Enthalpy |
|------|---------------|
| H-H  | 436 kJ/mol    |
| H-C  | 414 kJ/mol    |
| H-O  | 464 kJ/mol    |
| C-C  | 347 kJ/mol    |
| C=C  | 611 kJ/mol    |
| C-O  | 360 kJ/mol    |
| C=O  | 799 kJ/mol    |
| O-O  | 142 kJ/mol    |
| O=O  | 498 kJ/mol    |

$$\Delta H_{\text{rxn}}^\circ = \sum \Delta H_{\text{BE}}^\circ (\text{all bonds broken in reactants}) - \sum \Delta H_{\text{BE}}^\circ (\text{all bonds formed in products})$$

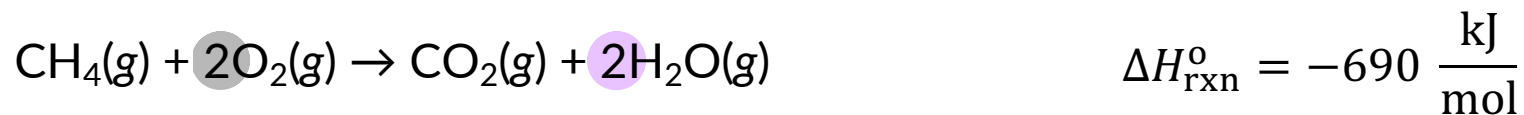
$$\begin{aligned} \Delta H_{\text{rxn}}^\circ &= \left[ 3 \times \Delta H_{\text{BE}}^\circ (\text{H}-\text{C}) + 1 \times \Delta H_{\text{BE}}^\circ (\text{C}=\text{C}) + 1 \times \Delta H_{\text{BE}}^\circ (\text{C}-\text{C}) + 1 \times \Delta H_{\text{BE}}^\circ (\text{C}=\text{O}) + 1 \times \Delta H_{\text{BE}}^\circ (\text{C}-\text{O}) + 1 \times \Delta H_{\text{BE}}^\circ (\text{H}-\text{O}) + 3 \times \Delta H_{\text{BE}}^\circ (\text{O}=\text{O}) \right] \\ &\quad - \left[ 3 \times 2 \times \Delta H_{\text{BE}}^\circ (\text{C}=\text{O}) + 2 \times 2 \times \Delta H_{\text{BE}}^\circ (\text{H}-\text{O}) \right] \\ &= \left[ 3 \times 414 \frac{\text{kJ}}{\text{mol}} + 1 \times 611 \frac{\text{kJ}}{\text{mol}} + 1 \times 347 \frac{\text{kJ}}{\text{mol}} + 1 \times 799 \frac{\text{kJ}}{\text{mol}} + 1 \times 360 \frac{\text{kJ}}{\text{mol}} + 1 \times 464 \frac{\text{kJ}}{\text{mol}} + 3 \times 498 \frac{\text{kJ}}{\text{mol}} \right] - \left[ 3 \times 2 \times 799 \frac{\text{kJ}}{\text{mol}} + 2 \times 2 \times 464 \frac{\text{kJ}}{\text{mol}} \right] \\ &= \left[ 5317 \frac{\text{kJ}}{\text{mol}} \right] - \left[ 6650 \frac{\text{kJ}}{\text{mol}} \right] \end{aligned}$$

$$\Delta H_{\text{rxn}}^\circ = -1333 \frac{\text{kJ}}{\text{mol}}$$

## EXAMPLE PROBLEM

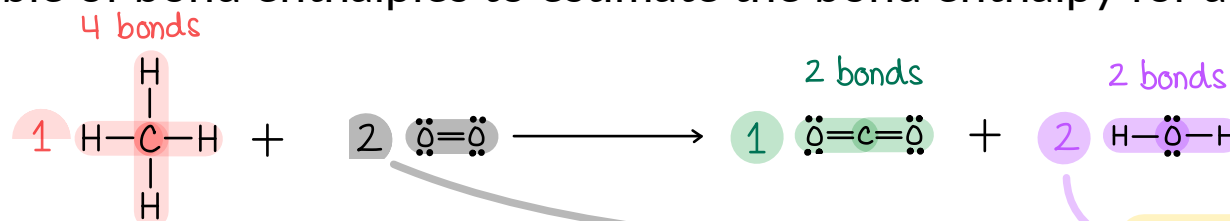
Learning Objective(s): 10.12

The thermochemical equation for the combustion of methane is given below.



Use this information and the table of bond enthalpies to estimate the bond enthalpy for a C–H bond.

| Bond | Bond Enthalpy                                     |
|------|---------------------------------------------------|
| C–H  | $\Delta H_{\text{BE}}^{\circ}(\text{H}-\text{C})$ |
| O=O  | 496 kJ/mol                                        |
| C=O  | 743 kJ/mol                                        |
| H–O  | 463 kJ/mol                                        |



$$\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{BE}}^{\circ} (\text{all bonds broken in reactants}) - \sum \Delta H_{\text{BE}}^{\circ} (\text{all bonds formed in products})$$

$$\Delta H_{\text{rxn}}^{\circ} = [4 \times \Delta H_{\text{BE}}^{\circ}(\text{H}-\text{C}) + 2 \times \Delta H_{\text{BE}}^{\circ}(\text{O}=\text{O})] - [1 \times 2 \times \Delta H_{\text{BE}}^{\circ}(\text{C}=\text{O}) + 2 \times 2 \times \Delta H_{\text{BE}}^{\circ}(\text{H}-\text{O})]$$

$$-690 \frac{\text{kJ}}{\text{mol}} = [4 \times \Delta H_{\text{BE}}^{\circ}(\text{H}-\text{C}) + 2 \times 496 \frac{\text{kJ}}{\text{mol}}] - [1 \times 2 \times 743 \frac{\text{kJ}}{\text{mol}} + 2 \times 2 \times 463 \frac{\text{kJ}}{\text{mol}}]$$

$$-690 \frac{\text{kJ}}{\text{mol}} = [4 \times \Delta H_{\text{BE}}^{\circ}(\text{H}-\text{C}) + 992 \frac{\text{kJ}}{\text{mol}}] - [3338 \frac{\text{kJ}}{\text{mol}}]$$

$$\Delta H_{\text{BE}}^{\circ}(\text{H}-\text{C}) = 414 \frac{\text{kJ}}{\text{mol}}$$

## BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Which elements could be X in the Lewis structure drawn? Select all that apply.

*Hint: There are only two correct answers.*

☐ C

☐ N

☐ P

☐ Ar

☐ As

